MULTIVARIABLE CALCULUS • UNIT 1.01

C Functions of Several Variables, Limits & Partial Derivatives

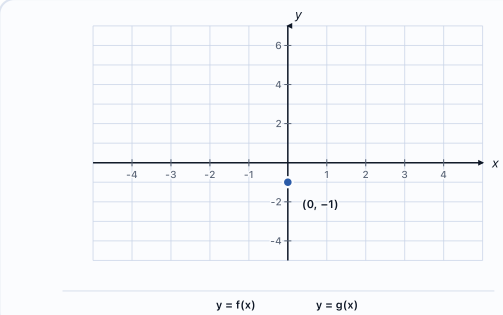
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Functions of Several Variables, Limits & Partial Derivatives Practice Exercises

1 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 10
B 13
C 26
D 52

2 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

A 12
B 18
C 6
D 24

3 MCQ

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

A -24
B 4
C 8
D -42

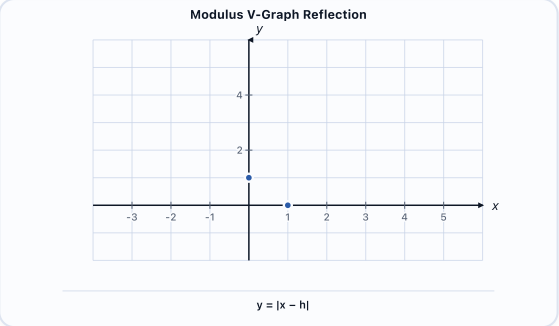
4 MCQ

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

A 2
B 4
C 6
D 0

5 MCQ

Let the linear scalar potential be $h(u, y) = 15u + 8y - 16$. Compute the squared gradient norm $\|\nabla h\|^2$.



A 289

B 293

C 281

D 304

6 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

A $\sqrt{14}$

B $2\sqrt{14}$

C $3\sqrt{14}$

D $4\sqrt{14}$

7 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

A 58

B 40

C 29

D 36

8 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

A $\langle 6, 8 \rangle$

B $\langle 8, 6 \rangle$

C $\langle 5, 6 \rangle$

D $\langle 6, 5 \rangle$

9 MCQ

A particle's position in three-dimensional space is given by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where t represents time. The velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t)$. Suppose the velocity components are related by the partial differential equations: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. If at time $t = 0$, the particle is at the origin $(0, 0, 0)$ and its velocity is $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, determine the value of $\frac{\partial^2 v_x}{\partial t^2}$ at $t = 0$.

Given Vector Components

Component x	Component y	Component z
0	0	0

A 16

B 26

C -16

D -26

10 MCQ

Consider the function $f(x, y) = 5x^3 - 2y^2 + 7$. Compute the value of the partial derivative $f_y(2, -3)$.

A -12

B 12

C -20

D 20

11 MCQ

Let $X = [0, 8] \times [0, 15]$ be a rectangle in the (u, y)-plane. Compute the double integral $\iint_X 44uy \, dA$.

A 158400

B 158404

C 158411

D 158376

12 MCQ

For $f(x, y) = 5x^2 + y^2$, find $f_y(2, -1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	5	1	6

A -2

B -1

C 20

D 4

13 MCQ

For $f(x, y) = 3x^2 + 6y^2$, find $f_x(-6, 1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	3	6	9

A 6

B -36

C -18

D 12

14 MCQ

For $f(x, y) = 8x^2 + 4y^2$, find $f_y(-5, 5)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	4	12

A -40

B 40

C 20

D -80

15 MCQ

For $f(x, y) = 5x^2 + 7y^2$, find $f_y(-5, 5)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	5	7	12

A -70

B 70

C 35

D -50

16 MCQ

For $f(x, y) = 8x^2 + 9y^2$, find $f_x(-4, -3)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	9	17

A -64

B -48

C -32

D -54

17 MCQ

For $f(x, y) = x^2 + 9y^2$, find $f_x(-3, -1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	9	10

A -2

B -3

C -6

D -18

18 MCQ

For $f(x, y) = x^2 + 3y^2$, find $f_y(2, 4)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	3	4

A 12

B 4

C -24

D 24

19 MCQ

For $f(x, y) = 8x^2 + 7y^2$, find $f_x(1, -4)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	7	15

A 16

B -64

C 8

D -56

20 MCQ

For $f(x, y) = 4x^2 + y^2$, find $f_y(-3, 4)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	1	5

A 8

B -6

C 4

D -24

21 MCQ

For $f(x, y) = 2x^2 + 7y^2$, find $f_x(-3, 2)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	7	9

A 8

B -6

C 28

D -12

22 MCQ

For $f(x, y) = 4x^2 + 7y^2$, find $f_x(-5, -2)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	7	11

A -16

B -20

C -40

D -28

23 MCQ

For $f(x, y) = 2x^2 + 3y^2$, find $f_x(-4, 3)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	3	5

A 12

B -8

C 18

D -16

24 MCQ

For $f(x, y) = 4x^2 + 7y^2$, find $f_y(-2, -6)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	7	11

A -28

B -84

C -42

D -16

25 MCQ

For $f(x, y) = 2x^2 + 6y^2$, find $f_y(5, 1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	6	8

A 60

B 12

C 6

D 20